STATEMENT OF WORK

1. Objective/Requirements

In support of high-temperature structural instrumentation research for hypersonic vehicles, NASA desires the ability to interrogate 100s of high-temperature rated fiber optic sensors. To achieve large sensor counts without purchasing multiple fiber optic data acquisition systems is to utilize Wavelength-Division Multiplexer (WDM) optical splitter boxes to double channel counts of existing fiber optic data acquisition systems. NASA has evaluated the performance of the WDM optical splitter boxes developed by Luna Innovations and has deem the performance acceptable to support a large-scale structural test.

2. Characteristics, Scope, and Specs

Specification of Wavelength-Division Multiplexer optical splitter boxes:

- Split wavelength domain of optical signal into two distinct domains:
  - Signal 1, pass optical signals below 1537nm
  - Signal 2, pass optical signals above 1542nm
  - Signal 3, combines outputs of Signal 1 and Signal 2
- Enclosure with 3 leads with LC/APC connectors
- Quantity: 90

3. Place of Performance

N/A

4. Period of Performance

6-10 weeks from purchase